

Inventory Management and Control Policy

Stock Level Optimization, Reorder Logic, Safety Stock Calculation & Velocity Categorization

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1. Objectives & Governance

This policy establishes enterprise standards for maintaining optimal stock levels, calculating replenishment reorder points, preventing stockouts, controlling carrying costs, and managing obsolete/dead inventory across SupplySense distribution networks.

2. Inventory Velocity Classification

Products are categorized based on sales velocity, average daily sales (ADS), and inventory movement thresholds:

Category	Average Daily Sales (ADS)	Movement Criteria	Policy Action
Fast Moving	> 50 units / day	Continuous daily sales movement	Maintain high safety stock; weekly replenishment
Normal Moving	10 – 50 units / day	Regular weekly sales movement	Standard reorder point replenishment
Slow Moving	< 10 units / day	Intermittent sales movement	Review monthly; reduce reorder quantities
Dead Stock	0 units / day	No sales movement for 90 consecutive days	Mandatory liquidation / supplier return review
Overstock	N/A	Available quantity > (Reorder Level × 3)	Freeze PO creation; initiate stock transfer
Stockout	N/A	Available quantity = 0 units	Trigger Emergency Procurement / Priority PO

3. Mathematical Reorder Point & Replenishment Logic

3.1 Reorder Point Formula: Inventory replenishment is evaluated continuously using the standardized reorder point mathematical formula:

Reorder Point (ROP) = (Expected Demand During Lead Time) + Safety Stock

Where: Expected Demand During Lead Time = (Average Daily Sales × Supplier Lead Time in Days).

3.2 Reorder Execution: When available_quantity (quantity_on_hand minus reserved_quantity) reaches or drops below the calculated ROP, an automated inventory risk alert is generated to trigger purchase requisition.

4. Safety Stock Calculation & Risk Factors

4.1 Standard Buffer: Safety stock is calculated to cushion against demand spikes and vendor delays.

4.2 Mandated Higher Safety Stock Drivers: Higher safety stock buffer (up to 50% increase) is mandatory under the following operational conditions:

- High Demand Volatility: Product categories with coefficient of variation > 30%.
- Long Supplier Lead Time: Lead times exceeding 30 calendar days.
- Supplier Unreliability: Supplier reliability score < 70% as per SSE-SUP-POL-001.
- Seasonal Spikes: Q4 festival peak season (Diwali / Year-end sales).
- Critical Revenue SKUs: Top 20 revenue-generating Laptops and Smartphones.

5. Inventory Valuation & Stock Adjustments

5.1 Valuation Method: Inventory is valued at Weighted Average Cost (WAC).

5.2 Write-off Thresholds: Stock write-offs due to damage, expiry, or theft under ₹25,000 require Inventory Lead approval. Write-offs exceeding ₹25,000 require CFO sign-off.